Lect# 2:
The Project Management Context and Processes

- **Theories & concept PM are not difficult to understand. What is difficult is implementing them.**
- **Project manager must consider many different components when managing project including:**
  1. **System View of PM**
  2. **Project phases & Project Life Cycle**
  3. **Understanding Organizations**
  4. **Suggested skills for PM**
  5. **PM Process Group**

1. A Systems View of Project Management

- To describe a more analytical approach to management and problem solving
- Three parts include:
  1. **Systems philosophy:** View things as systems; interacting components working within an environment to fulfill some purpose
     - Ex. Human body system with sub systems– brain, circulatory system, & digestive system.
  2. **Systems analysis:** problem-solving approach that;
     - Define scope of sys to be studied,
     - Divide it into component parts,&
     - Identify & evaluate its problem, opportunities constraints & needs.
  3. **Systems management:** Addresses business, technological, and organizational issues before making changes to systems

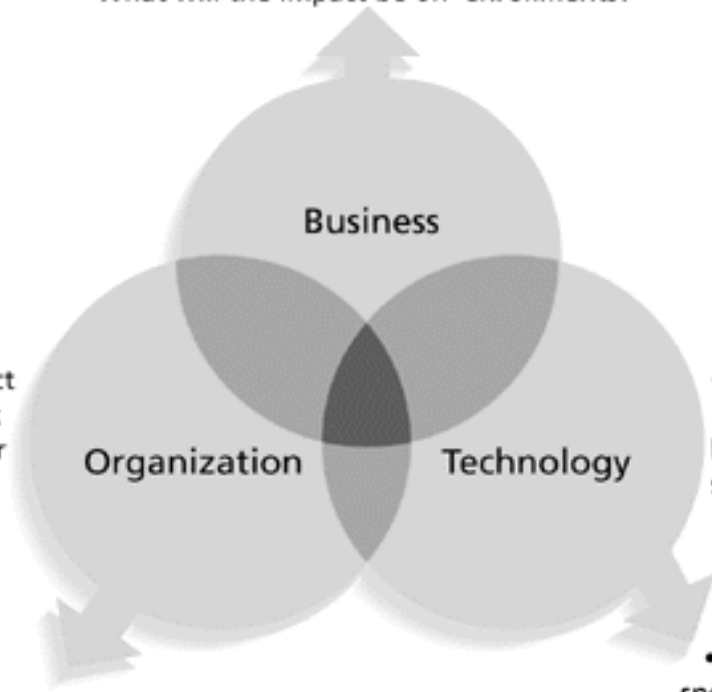
Fig. 2-1. Three Sphere Model for Systems Management

•What will the laptop project cost the college?

•What will it cost students?

•What will support costs be?

•What will the impact be on enrollments?



•Will the laptop project affect *all* students, just traditional students, or only certain majors?

•How will the project affect students who already have PCs or laptops?

•Who will train students, faculty, and staff?

•Who will administer and support training?

•Should the laptops use Macintosh, Windows, or both types of operating systems?

•What applications software will be loaded?

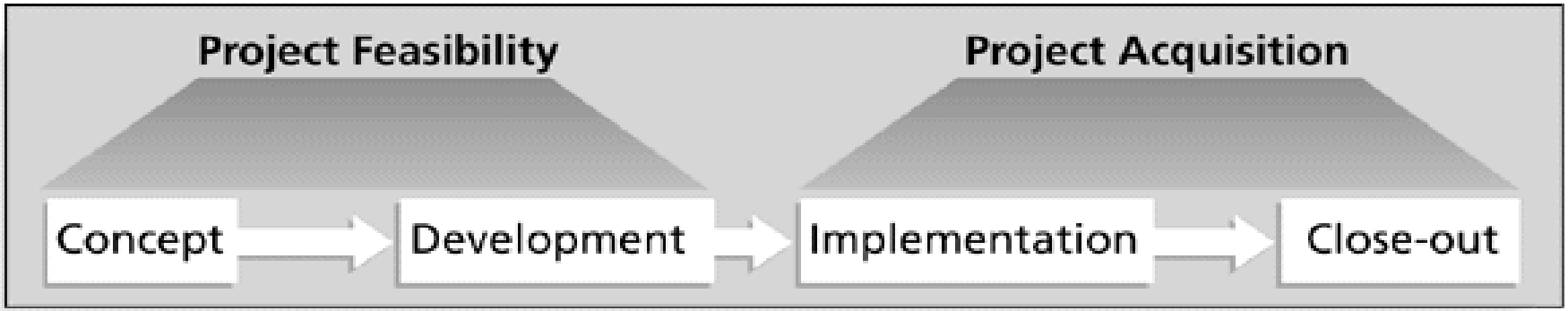
•What will the hardware specifications be?

•How will the hardware impact LAN and Internet access?

2. Project Phases and the Project Life Cycle

- A project life cycle is a collection of project phases
- Project phases vary by project or industry, but some general phases include
  - concept
  - development
  - implementation
  - close-out
- A project must successfully complete each phase before moving on to the next

Phases of the Project Life Cycle



Management plan	Project plan	Last work package	Completed work
Preliminary cost estimate	Budgetary cost estimate	Definitive cost estimate	Lessons learned
3-level WBS	6+-level WBS	Bulk of time should be spent here	Customer acceptance

Product Life Cycles

- Products also have life cycles
- The Systems Development Life Cycle (SDLC)

  a framework for describing the phases involved in developing and maintaining information systems
- Typical SDLC phases include:

  planning, analysis, design, implementation, and support
- Sample SDLC Models
  - Waterfall model: has well-defined, linear stages of systems development and support
  - Spiral model: shows that software is developed using an iterative or spiral approach rather than a linear approach
  - Incremental release model: provides for progressive development of operational software
  - Prototyping model: used for developing prototypes to clarify user requirements

Distinguishing Project Life Cycles and *Product* Life Cycles

- **The project life cycle applies to all projects, regardless of the products being produced**
- ***Product* life cycle models vary considerably based on the nature of the *product***
- ***Most large IT products are developed as a series of projects (see next slid)***
- **Project management is a cross life cycle activity done in all of the *product* life cycle phases**
- **Since project management needs to occur during all phases of systems development life cycle, critical for IT pro to understand and practice good project management .**

Example: Product & Project

- In system planning phase for new IS, can include a project to hire consulting firm to help identity and evaluate new order system or new ledger system.
- In system analysis phase, include project to create process model for business functions or to create data model
- In implementation phase, a project to hire contract programmer to code

Importance of Project Phases and Management Reviews.

- A project should successfully pass through each of the project phases in order to continue on to the next
- Management reviews (also called phase exits or kill points) should occur after each phase to evaluate the project's progress, likely success, and continued compatibility with organizational goals

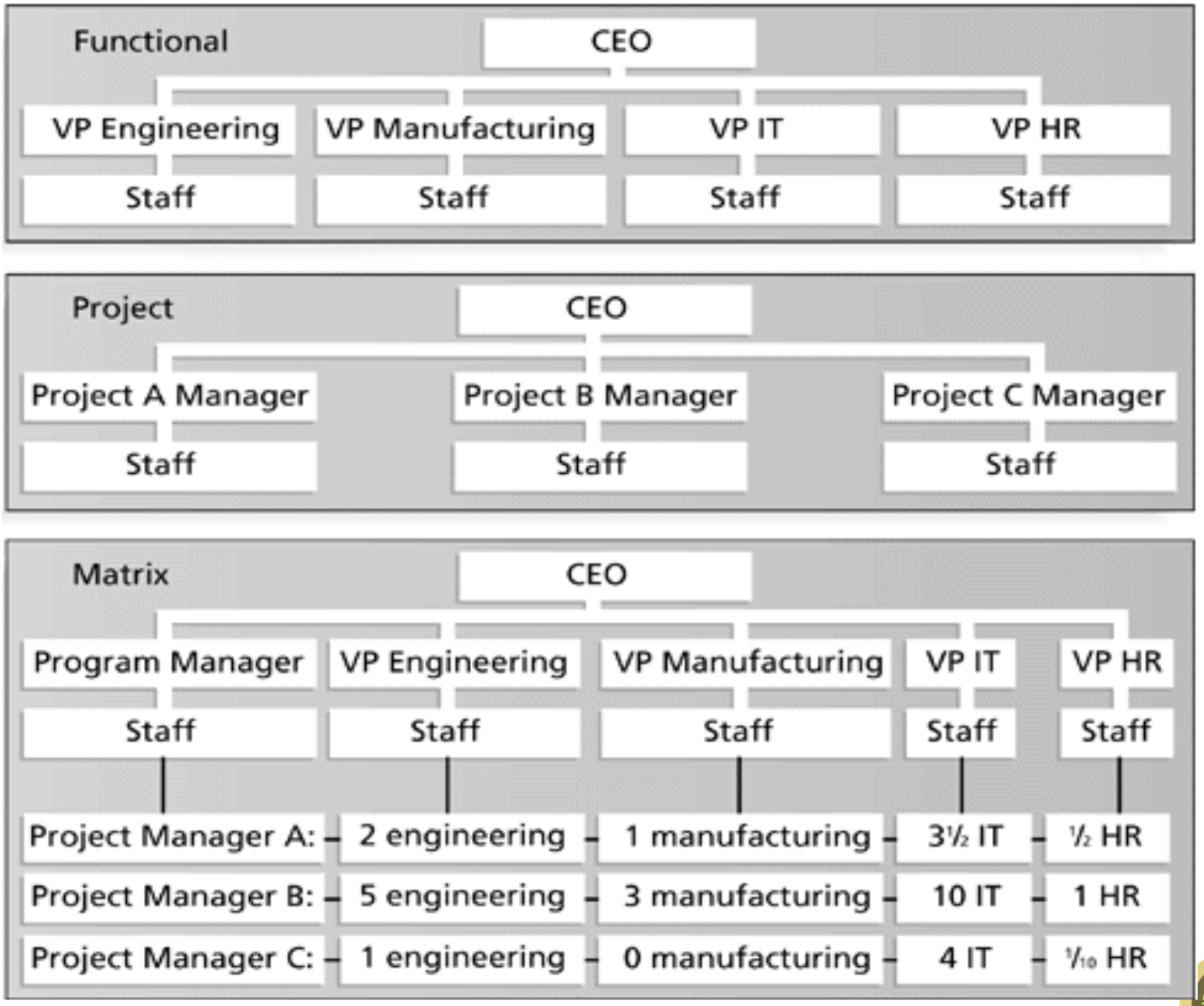
3. Understanding Organizations

Structural frame: Focuses on roles and responsibilities, coordination and control. Organization charts help define this frame.	**Human resources frame:** Focuses on providing harmony between needs of the organization and needs of people.
Political frame: Assumes organizations are coalitions composed of varied individuals and interest groups. Conflict and power are key issues.	**Symbolic frame:** Focuses on symbols and meanings related to events. Culture is important.

Many Organizations Focus on the Structural Frame

- Most people understand what organizational charts are
- Many new managers try to change organizational structure when other changes are needed
- 3 basic organization structures
  - functional
  - project
  - matrix

Functional, Project, and Matrix Organizational Structures



Recognize the Importance of Project Stakeholders

- Recall that project stakeholders are the people involved in or affected by project activities
- Project managers must take time to identify, understand, and manage relationships with all project stakeholders
- Using the four frames of organizations can help meet stakeholder needs and expectations

Table 2-2. Fifteen Project Management Job Functions

1. Define scope of project
2. Identify stakeholders, decision-makers, and escalation procedures
3. Develop detailed task list (work breakdown structures)
4. Estimate time requirements
5. Develop initial project management flow chart
6. Identify required resources and budget
7. Evaluate project requirements
8. Identify and evaluate risks
9. Prepare contingency plan
10. Identify interdependencies
11. Identify and track critical milestones
12. Participate in project phase review
13. Secure needed resources
14. Manage the change control process
15. Report project status

4. Suggested Skills for a Project Manager

- Communication skills: listening, persuading
- Organizational skills: planning, goal-setting, analyzing
- Team Building skills: empathy, motivation, esprit de corps
- Leadership skills: sets example, energetic, vision (big picture), delegates, positive
- Coping skills: flexibility, creativity, patience, persistence
- Technological skills: experience, project knowledge

5. Project Management Process Groups

- Project management can be viewed as a number of interlinked processes
- The project management process groups include
  1. initiating processes
  2. planning processes
  3. executing processes
  4. controlling processes
  5. closing processes

Figure 2-5. Level of Process Group Activity Over Time

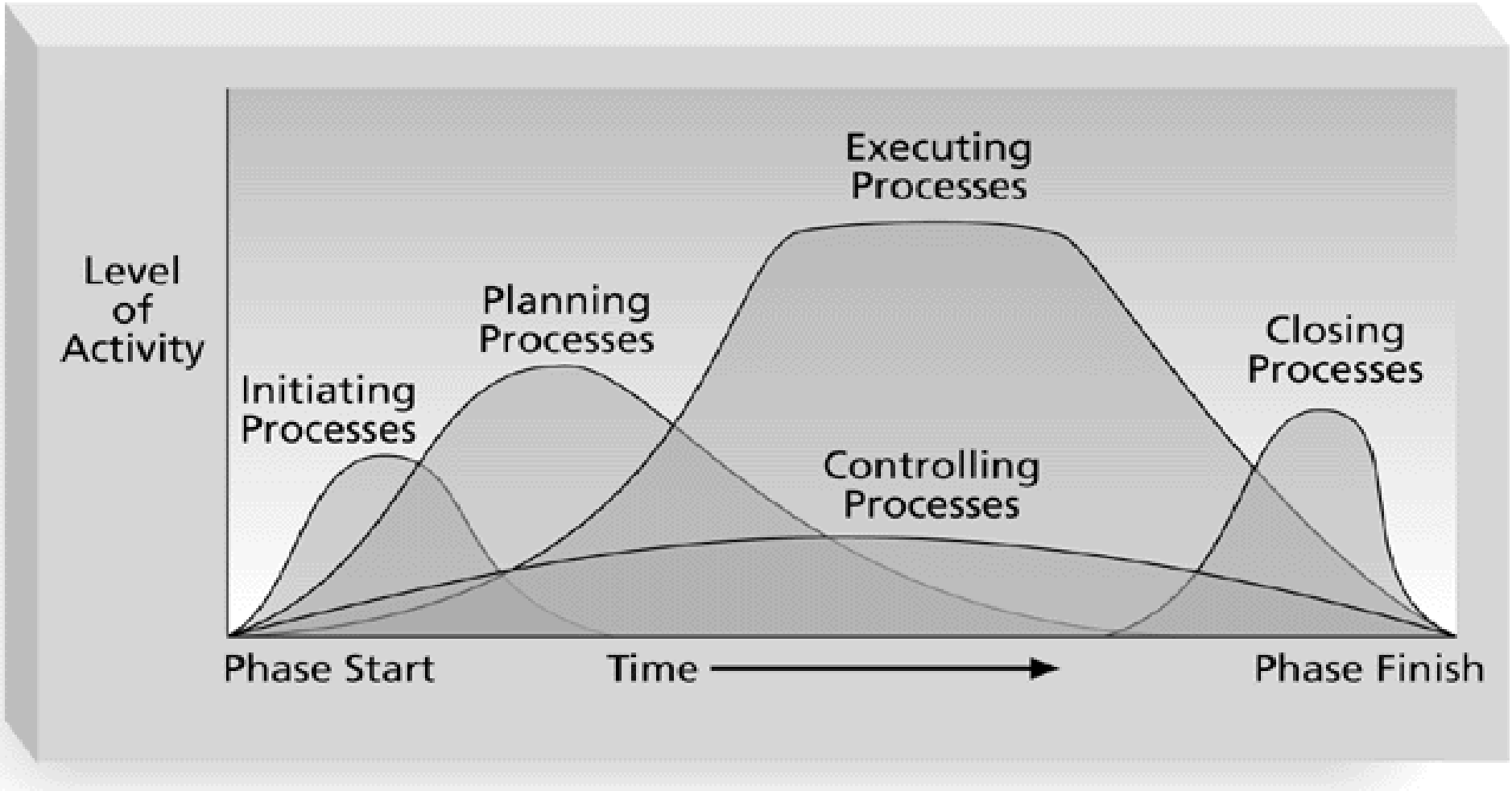


Table 2-4. Relationships Among Project Process Groups, Activities, and Knowledge Areas

	Project Process Groups				
Knowledge Area	Initiating	Planning	Executing	Controlling	Closing
Integration		Project plan development	Project plan execution	Overall change control	
Scope	Initiation	Scope planning	Scope verification	Scope change control	
Time		Activity definition		Schedule control	
		Activity sequencing			
		Activity duration estimating			
		Schedule development			
Cost		Resource planning		Cost control	
		Cost estimating			
		Cost budgeting			
Quality		Quality planning	Quality assurance	Quality control	
Human resources		Organizational planning	Team development		
		Staff acquisition			
Communications		Communications planning	Information distribution	Performance reporting	Administrative closure
Risk		Risk identification		Risk response control	
		Risk quantification			
		Risk response development			
Procurement		Procurement planning	Solicitation		Contract close-out
		solicitation planning	Source selection		
			Contract administration		